Mathematical Statistics I
HOMEWORK 9

Kayla M. Dyer

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1.

a.

Y_2/Y_1	0	1
0	0.38	0.17
1	0.14	0.02
2	0.24	0.05

**Marginal Probability for Y_1 and Y_2

$$P(Y_1 = 0) = 0.76$$

$$P(Y_1 = 1) = 0.24$$

$$P(Y_2 = 0) = 0.55$$

$$P(Y_2 = 1) = 0.16$$

$$P(Y_2 = 2) = 0.29$$

$$P(Y_2 = 0 \mid Y_1 = 0) = \frac{P(Y_2 = 0, Y_1 = 0)}{P(Y_1 = 0)} = \frac{0.38}{0.76} = \frac{1}{2} = 0.50$$

$$P(Y_2 = 1 \mid Y_1 = 0) = \frac{P(Y_2 = 1, Y_1 = 0)}{P(Y_1 = 0)} = \frac{0.14}{0.76} \sim 0.1842$$

$$P(Y_2 = 2 \mid Y_1 = 0) = \frac{P(Y_2 = 2, Y_1 = 0)}{P(Y_1 = 0)} = \frac{0.24}{0.76} \sim 0.3158$$

Conditional probability of Y_1 given that $Y_2 = 0$

$$P(Y_2 = 0 \mid Y_1 = 0) = 0.50$$

$$P(Y_2 = 1 \mid Y_1 = 0) \sim 0.1842$$

$$P(Y_2 = 2 \mid Y_1 = 0) \sim 0.3158$$

$$0.50 + 0.1842 + 0.3158 = 1$$

Yes, a legit probability distribution.

b.

Y_2/Y_1	0	1
0	0.38	0.17
1	0.14	0.02
2	0.24	0.05

**Marginal Probability for Y_1 and Y_2

$$P(Y_1 = 0) = 0.76$$

$$P(Y_1 = 1) = 0.24$$

$$P(Y_2 = 0) = 0.55$$

$$P(Y_2 = 1) = 0.16$$

$$P(Y_2 = 2) = 0.29$$

$$P(Y_1 = 0 \mid Y_1 = 2) = \frac{P(Y_1 = 0, Y_2 = 2)}{P(Y_2 = 2)} = \frac{0.24}{0.29} \sim 0.8276$$

$$P(Y_1 = 0 \mid Y_1 = 2) \sim 0.8276$$

The probability that a child survived given that they were in a car-seat belt is about 82.76%.

2.

a.

$$f(x, y) = e^{-(x+y)}$$

$$f_X(x) = e^{-x}$$

$$f_Y(y) = e^{-y}$$

$$f_{X|Y}(x|y) = \frac{e^{-(x+y)}}{e^{-y}} = e^{-x}, \qquad x > 0$$

$$\int_0^\infty f_{X|Y}(x|y)dx = \int_0^\infty e^{-x}dx = 1$$

$$f_{X|Y}(x|y) = e^{-x}, \qquad x > 0$$

b.

$$f_{X|Y}(x|y) = e^{-x}$$

$$f_X(x) = e^{-x}$$

$$f_{X|Y}(x|y) = f_X(x)$$

$$e^{-x} = e^{-x}$$

The conditional density of X given $Y = y$ is the same as the marginal density of X. This implies that the two random variables, X and Y are independent.

3.

a.

$$f_X(x) = \int_0^1 f(x,y)dy = \int_0^1 (x+y)dy = \left[xy + \frac{y^2}{2}\right]_0^1 = x + \frac{1}{2}$$

$$f_Y(y) = \int_0^1 f(x,y)dx = \int_0^1 (x+y)dx = \left[\frac{x^2}{2} + yx\right]_0^1 = y + \frac{1}{2}$$

$$f_X(x)f_Y(y) = \left(x + \frac{1}{2}\right)\left(y + \frac{1}{2}\right) = xy + \frac{1}{2}x + \frac{1}{2}y + \frac{1}{4}$$

$$f(x,y) = x + y$$

$$f_X(x)f_Y(y) \neq f(x,y)$$

$$\left(x + \frac{1}{2}\right)\left(y + \frac{1}{2}\right) = xy + \frac{1}{2}x + \frac{1}{2}y + \frac{1}{4} \neq x + y$$

$$f_X(x) = x + \frac{1}{2}$$

$$f_Y(y) = y + \frac{1}{2}$$

They are not equal. Since $f(x,y) \neq f_X(x) f_Y(y)$, the random variables X and Y are not independent.

b.

$$E(Z) = 30E(X) + 25E(Y)$$

$$f_X(x) = x + \tfrac{1}{2},\ f_Y(y) = y + \tfrac{1}{2}\ , \qquad 0 \leq x, \quad y \leq 1$$

$$E(X) = \int_0^1 x f_X(x) dx = \int_0^1 x\left(x + \frac{1}{2}\right) dx = \int_0^1 \left(x^2 + \frac{1}{2}x\right) dx = \left[\frac{x^3}{3} + \frac{x^2}{4}\right]_0^1 = \frac{1}{3} + \frac{1}{4}$$

$$E(X) = \frac{7}{12} \sim 0.5833$$

$$E(Z) = 30E(X) + 25E(Y) = 30\left(\frac{7}{12}\right) + 25\left(\frac{7}{12}\right) = \frac{385}{12} \sim 32.0833$$

$$E(30X + 25Y) = \frac{385}{12} \sim 32.0833$$

The expected value of this measure of productivity is about 32.0833.

4.

Y_1/Y_2	0	1	2	3
0	0	0.035714	0.071429	0.011905
1	0.047619	0.285714	0.142857	0
2	0.142857	0.214286	0	0
3	0.047619	0	0	0

Y_1	Row values	Sum$f_{Y1}(y_1)$
0	$0 + 0.035174 + 0.071429 + 0.011905$	0.119048
1	$0.047619 + 0.285714 + 0.142857 + 0$	0.476190
2	$0.142857 + 0.214286 + 0 + 0$	0.357143
3	$0.47619 + 0 + 0 + 0$	0.047619

Y_2	Column values	Sum $f_{Y2}(y_2)$
0	$0 + 0.047619 + 0.142857 + 0.047619$	0.238095
1	$0.035714 + 0.285714 + 0.214286 + 0$	0.535714
2	$0.071429 + 0.142857 + 0 + 0$	0.214286
3	$0.011905 + 0 + 0 + 0$	0.011905

$E(Y_1) = (0)(0.119048) + (1)(\ 0.476190) + (2)(\ 0.357143) + (3)(0.047619)$

$$E(Y_1) \sim 1.333$$

$$E(Y_2) = (0)(0.238095) + (1)(0.535714) + (2)(0.214286) + (3)(0.011905)$$

$$E(Y_2) = 1.000$$

** Any cell where $Y_1 = 0$ or $Y_2 = 0$ ignore because product$= 0$.

$$E(Y_1Y_2) = (1 \cdot 1 \cdot 0.285714) + (1 \cdot 2 \cdot 0.142857) + (2 \cdot 1 \cdot 0.214286)$$

$$E(Y_1Y_2) = 1.000$$

$$Cov(Y_1Y_2) = E(Y_1Y_2) - E(Y_1)E(Y_2)$$

$$Cov(Y_1Y_2) = 1.000 - (1.333)(1.000)$$

$$Cov(Y_1Y_2) = -0.333$$

5.

a.

$$f_{XY}(x, y) = 16xe^{-2(x+2y)} \quad , \qquad x > 0, y > 0$$

$$f_Y(y) = \int_0^\infty f_{XY}(x, y)dx = \int_0^\infty 16xe^{-2(x+2y)}dx = 16xe^{-4y} \int_0^\infty xe^{-2x}\,dx$$

$$f_Y(y) = \int_0^\infty xe^{-2x}dx = \frac{1}{2^2}\Gamma(2) = \frac{1}{4}(1!) = \frac{1}{4}$$

$$f_Y(y) = 16xe^{-4y}\left(\frac{1}{4}\right) = 4e^{-4y}$$

$$f_Y(y) = 4e^{-4y}$$

$$Y \sim Exp(\lambda = 4) \ \ or. Exp\left(\beta = \frac{1}{4}\right)$$

$$f_X(x) = \int_0^{\infty} f_{XY}(x,y)dx = \int_0^{\infty} 16xe^{-2(x+2y)}dy = 16xe^{-2x}\int_0^{\infty} e^{-4y}\,dy$$

$$f_X(x) = 16xe^{-2x}\left(\frac{1}{4}\right) = 4xe^{-2x}$$

$$f_X(x) = \frac{1}{\Gamma(2)(\beta^2)}x^{2-1}e^{-\frac{x}{\beta}} = \frac{1}{\Gamma(2)\left(\frac{1}{2}\right)^2} = 4xe^{-2x}$$

$$f_X(x) = 4xe^{-2x}$$

$$X \sim Gamma\left(\alpha = 2, \beta = \frac{1}{2}\right)$$

$$f_{XY}(x,y) = f_X(x)f_Y(y)$$

$$f_X(x)f_Y(y) = (4xe^{-2x})(4e^{-4y}) = 16xe^{-2(x+2y)} = f_{XY}(x,y)$$

$$f_X(x)f_Y(y) = f_{XY}(x,y)$$

They are equal so $X \ and \ Y$ are independent.

b.

$$E(X) = \int_0^{\infty} xf_X(x)dx = \int_0^{\infty} 4x^2e^{-2x}dx = 4 \cdot \frac{\Gamma(3)}{2^3} = 4 \cdot \frac{2!}{8} = 1$$

$$E(Y) = \int_0^{\infty} yf_Y(y)dy = \int_0^{\infty} 4ye^{-4y}dy = 4 \cdot \frac{\Gamma(2)}{4^2} = 4 \cdot \frac{1!}{16} = \frac{1}{4}$$

$$E(XY) = E(X)E(Y) = 1 \cdot \frac{1}{4} = \frac{1}{4} = 0.25$$

$$E(XY) = 0.25$$

c.

$$Cov(X,Y) = E(XY) - E(X)E(Y)$$

$$Cov(X,Y) = \frac{1}{4} - (1)\left(\frac{1}{4}\right) = 0$$

$$Cov(X,Y) = 0$$

Independence/ Zero Covariance.